ISE552 · Artificial Intelligence

Unit 1: Introduction to AI, Intelligent Agents & Problem Solving

Handwritten Student Notebook

Syllabus Topics Covered:

OFFICIAL SYLLABUS TOPICS COVERED Definitions of AI: Thinking humanly (cognitive modeling), acting humanly (Turing test), thinking rationally (laws of thought), and acting rationally (rational agent approach). Foundations of AI: Philosophy, Mathematics, Economics, Neuroscien...

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Handwritten-style notes (generated from unit notes) - verify with your textbook and faculty notes.

What is AI? The Four Historical Approaches

Source: Russell & Norvig 4e, Ch. 1.1, pp. 1–5; AI-Intelligent_Agents.pdf slides 1–15

§ 1

Historically, researchers have approached **Artificial Intelligence** along two fundamental dimensions: whether the goal is to match human performance or adhere to an ideal standard of rationality, and whether the focus is on internal thought processes or external behavior (actions).

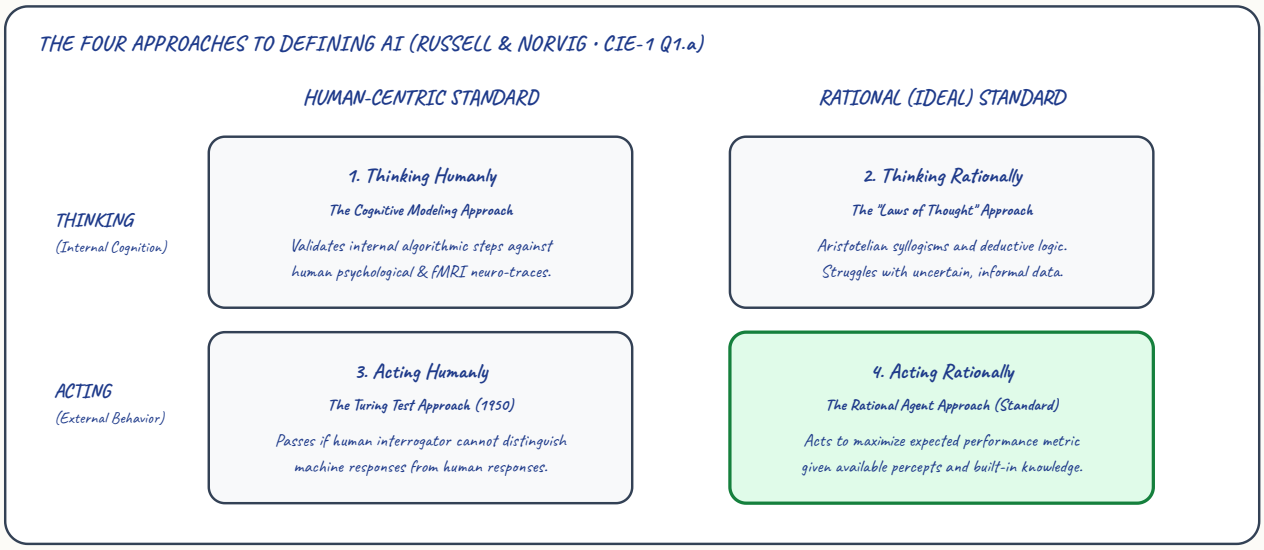


Figure 1.1: The Four Approaches to Defining Artificial Intelligence (Russell & Norvig · Ramaiah CIE-1 Q1.a)

*Modern AI adopts the **Acting Rationally** (Rational Agent) standard, as it provides a general, mathematically rigorous framework independent of human biological constraints.*

Focus Dimension	Human-Centric Approach	Rationality-Centric (Ideal Standard)
Thinking (Mind / Reasoning)	**1. Thinking Humanly (Cognitive Modeling)** Requires determining how humans think through psychological experiments, brain scans (fMRI), or introspective analysis. Compares computer traces to human cognitive behavior.	**2. Thinking Rationally (Laws of Thought)** Rooted in Aristotle's syllogisms and formal mathematical logic. Focuses on irrefutable reasoning patterns ($P \implies Q, P \vdash Q$). Limited by computational tractability and expressing informal uncertain knowledge.
Acting (Behavior / Output)	**3. Acting Humanly (The Turing Test)** Proposed by Alan Turing (1950). An agent passes if a human interrogator cannot distinguish its conversational replies from those of a human. Requires NLP, knowledge representation, automated reasoning, and machine learning.	**4. Acting Rationally (The Rational Agent) — MODERN STANDARD** A **rational agent** is one that acts so as to achieve the best expected outcome, or best expected utility when uncertainty is present. Correct inference is one way to act rationally, but reflexive safety actions are also rational.

WHY MODERN AI FOCUSES ON THE RATIONAL AGENT APPROACH:

The rational agent approach is superior to human modeling for two main reasons:

1. It is more general than the "laws of thought" approach, because correct inference is only one of several mechanisms for achieving rationality (e.g. reflex actions can be rational without logical deduction).
2. It is more mathematically amenable to scientific development than approaches based on human behavior, because the standard of rationality is mathematically well-defined and completely independent of human biological quirks and cognitive errors.

SOURCE: UNIT-1.PDF (SLIDES 37–49) · FOUNDATIONAL TAXONOMY & EVOLUTION:
The following subsections provide the core academic distinctions, evolutionary milestones, and taxonomy detailed in the professor's latest lecture deck.

AI vs Traditional Programming & Natural and Artificial Intelligence

A fundamental examination and interview topic is understanding how Artificial Intelligence fundamentally departs from classic algorithmic software:

Dimension	Traditional Programming	Artificial Intelligence (AI)
Core Mechanism	Rule-driven: Programmer writes explicit if-else logic and hardcoded algorithms.	Data & Model-driven: Systems learn patterns from data and infer decisions.
Adaptability	Static: Cannot handle novel inputs without explicit manual code modification.	Adaptive: Generalizes to unseen inputs through learning and probabilistic reasoning.
Output Nature	Deterministic: Identical input always yields identical exact output.	Probabilistic / Heuristic: Yields optimal or satisficing solutions under uncertainty.
Problem Suitability	Structured, arithmetic, and repetitive payroll/database tasks.	Perceptual tasks (vision, speech, reasoning, autonomous control, game playing).

Natural Intelligence vs Artificial Intelligence

Natural Intelligence arises from billions of interconnected biological neurons developed over millions of years of evolution, exhibiting emotional cognition, moral reasoning, and extreme energy efficiency (~20 Watts in human brain). **Artificial Intelligence** uses silicon processors, matrix arithmetic, and mathematical optimization models, capable of processing massive data volumes at millisecond latency without fatigue, but lacking intrinsic consciousness and subjective experience.

Evolution and History of AI

The historical trajectory of AI is characterized by alternating phases of high optimism and "AI Winters":

- **1950 — Turing's Foundations:** Alan Turing publishes Computing Machinery and Intelligence, formulating the Imitation Game (Turing Test).
- **1956 — The Birth of AI:** John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon organize the **Dartmouth Summer Research Project**, where McCarthy coins the phrase "Artificial Intelligence".
- **1970s–1980s — AI Winters:** Early neural network and translation projects faced severe computational bottlenecks and combinatorial explosion, leading to cancelled government funding and the famous "AI Winters".
- **1980s — Expert Systems Boom:** Rise of commercial rule-based knowledge systems (e.g. R1/XCON at DEC).
- **2000s–2020s — Deep Learning Revolution:** Driven by massive digital datasets (Big Data), specialized GPU/TPU parallel hardware, and deep neural network architectures (AlexNet, Transformers, LLMs).

Foundations of AI: The 8 Interdisciplinary Pillars

Artificial Intelligence builds upon key foundational disciplines (Russell & Norvig Ch. 1.2; Unit-1.pdf slides 43–45):

Discipline	Foundational Contribution to AI
Philosophy	Can formal rules be used to draw valid conclusions? Where does knowledge come from? (Mind-body dualism, materialism, empiricism).
Mathematics	Formal logic, probability theory (Bayes rule), algorithms, and computability theory (Turing machines, NP-completeness).
Economics	Decision theory, game theory (von Neumann & Morgenstern), utility functions, and satisficing behavior (Herbert Simon).
Neuroscience	How the human brain processes information, neuronal firing mechanisms, and biological neural architectures.
Psychology	Cognitive science: How do humans perceive, remember, and process conceptual knowledge?
Computer Engineering	High-performance digital computing, parallel architectures, microprocessors, and specialized accelerators (GPUs, TPUs).
Control Theory	Cybernetics, feedback loops, state estimation, and designing stable systems that minimize error over time.

Discipline	Foundational Contribution to AI
Linguistics	How language relates to thought, formal grammars, syntax, semantics, and Natural Language Processing (Chomsky).

Taxonomy of AI: Capabilities vs Functionalities

In standard academic categorization (Unit-1.pdf slides 46-47), AI is classified into two distinct dimensions:

Classification Basis	Level / Category	Definition & Real-World Example
Based on Capabilities	**Narrow AI (ANI / Weak AI)**	Specialized to perform a single dedicated task efficiently (e.g. Siri, AlphaGo, Spam filters, Medical diagnosis). Current state of AI.
	General AI (AGI / Strong AI)	Hypothetical machine with human-level cognitive capabilities across any intellectual task. Can learn, adapt, and reason generally.
	Super AI (ASI)	Hypothetical intelligence far surpassing the smartest human brains in scientific creativity, wisdom, and social skills.
Based on Functionality	**1. Reactive Machines**	No memory of past events; perceives the present state and takes the best action immediately (e.g. IBM Deep Blue chess machine).
	2. Limited Memory	Stores past observations for a short period to inform current decisions (e.g. Autonomous self-driving vehicles, LLM context windows).
	3. Theory of Mind	Future AI capable of understanding that other agents have thoughts, emotions, and beliefs that influence their behavior.
	4. Self-Aware AI	Autonomous consciousness and self-awareness; understands its own internal state, emotions, and existence (theoretical).

QUICK RECALL

- 4 Approaches: Thinking Humanly, Acting Humanly (Turing), Thinking Rationally (Logic), Acting Rationally (Agents).
- Turing Test requires: NLP, Knowledge Representation, Automated Reasoning, Machine Learning. (Total Turing Test adds Computer Vision & Robotics).
- Modern AI standard: ***Acting Rationally*** (maximizing expected performance).
- AI vs Traditional: Rule-driven & static vs. Data-driven & adaptive.
- AI Types: Narrow (ANI), General (AGI), Super (ASI) • Reactive, Limited Memory, Theory of Mind, Self-Aware.

Intelligent Agents and Environments

Source: Russell & Norvig 4e, Ch. 1.1, pp. 2–4; Turing (1950); slides 16–22

§ 2

An ***agent*** is anything that can be viewed as perceiving its ***environment*** through ***sensors*** and acting upon that environment through ***actuators***.

AGENT MATHEMATICAL FORMALISM:

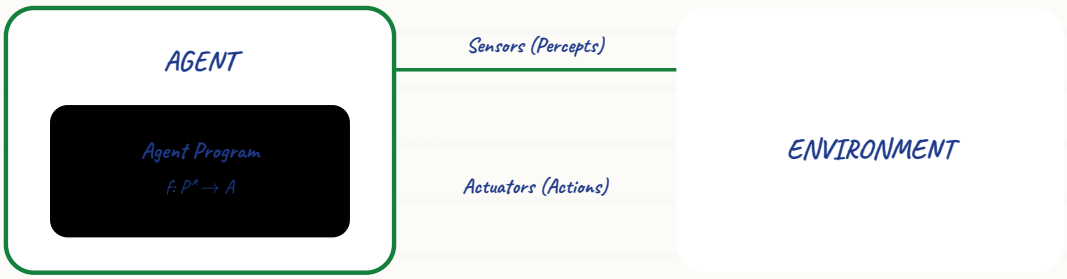
- **Percept:** The agent's perceptual inputs at any given instant.
- **Percept Sequence:** The complete history of everything the agent has ever perceived: P^*.
- **Agent Function:** An abstract mathematical mapping from any percept sequence to an action:

$$f : P^* \to A$$

- **Agent Program:** The concrete computational implementation of the agent function running on a physical computing architecture:

$$\text{Agent} = \text{Architecture} + \text{Program}$$

Agent Interaction with its Environment



QUICK RECALL

- Agent perceives environment via sensors; affects environment via actuators.
- Agent Function is an abstract mathematical mapping: $f : P^* \rightarrow A$.
- Agent Program is the concrete software code running on the physical architecture.

The Concept of Rationality and Autonomy

Source: Russell & Norvig 4e, Ch. 1.2, pp. 5–16; slides 23–30

§ 3

What is rational at any given time depends on four factors: the ***performance measure*** that defines the criterion of success, the agent's ***prior knowledge*** of the environment, the ***actions*** that the agent can perform, and the agent's ***percept sequence*** to date.

> **DEFINITION OF A RATIONAL AGENT:**
> For each possible percept sequence, a rational agent should select an action that is expected to ***maximize its performance measure***, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Critical Distinctions in Rationality

Concept	Key Definition	Crucial Exam Contrast
Omniscience vs Rationality	An **omniscient agent** knows the actual outcome of its actions and can act accordingly.	Omniscience is impossible in reality. Rationality maximizes **expected performance** based on available information, not actual outcome (e.g. looking both ways before crossing a street is rational, even if a meteorite strikes).
Information Gathering & Exploration	Doing actions to modify future percepts (e.g. looking around, medical tests).	A rational agent must gather information and explore unfamiliar environments rather than blindly acting on initial assumptions.
Autonomy	An agent is **autonomous** to the extent that its behavior is determined by its own experience (learning), rather than solely by the initial knowledge programmed by its designer.	A non-autonomous agent relies entirely on pre-baked assumptions and fails when the environment changes. Truly rational agents learn to become autonomous over time.

QUICK RECALL

- Rationality depends on: Performance measure, Prior knowledge, Actions, Percept sequence.
- Rationality ≠ Omniscience. Rationality maximizes expected success, not perfection in hindsight.
- Autonomy = learning from experience rather than relying solely on built-in prior knowledge.

The PEAS Specification Framework

Source: Russell & Norvig 4e, Ch. 1.3, pp. 16–28; slides 31–45

§ 4

To design a rational agent, we must first specify the **task environment** using the **PEAS** framework: **P**erformance measure, **E**nvironment, **A**ctuators, and **S**ensors.

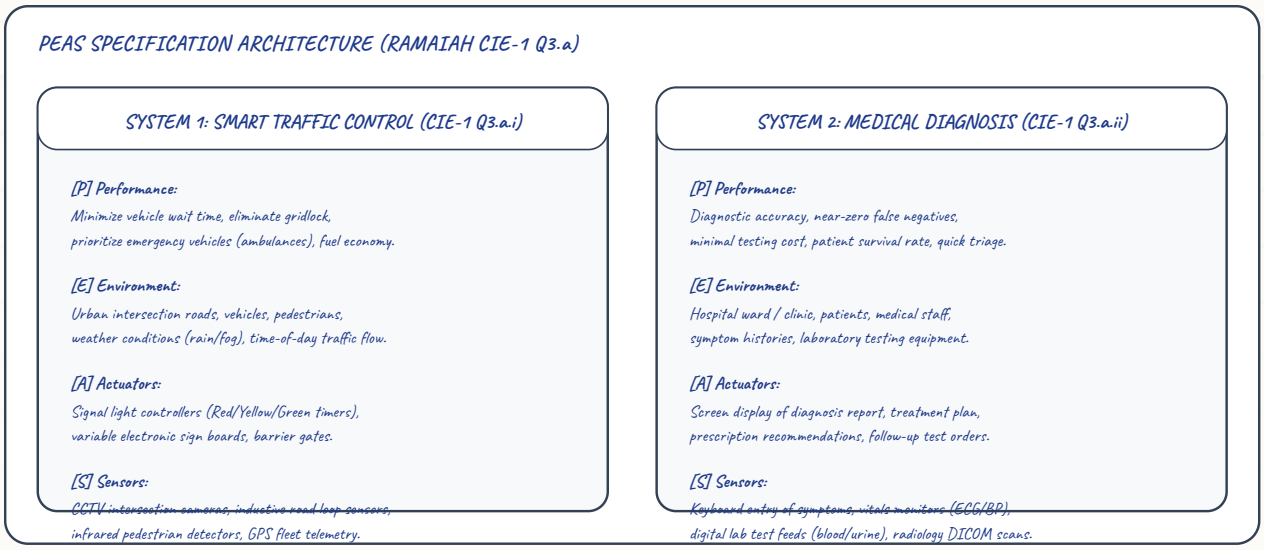


Figure 1.3: PEAS Specification Architecture for Smart Traffic Control & Medical Diagnosis (Ramaiah CIE-1 Q3.a)

Specifying Performance (P), Environment (E), Actuators (A), and Sensors (S) is the mandatory first step before selecting an agent architecture.

Agent Type	Performance Measure (P)	Environment (E)	Actuators (A)	Sensors (S)
Automated Taxi Driver	Safe, fast, legal trip; maximize profits; minimize passenger discomfort; fuel efficiency.	City roads, traffic, pedestrians, weather conditions, customers.	Steering wheel, accelerator, brakes, gear shift, horn, turn signals, display screen.	Video cameras, sonar/radar, LIDAR, GPS, speedometer, engine sensors, accelerometer.
Medical Diagnosis System	Healthy patient, minimized healthcare costs, zero misdiagnoses, patient recovery time.	Patient, hospital clinical staff, insurance systems, laboratory.	Display screen (diagnostic reports, treatment plans, prescription suggestions).	Keyboard (symptoms entered by doctor), patient medical history, lab test results.
Part-Picking Robot	Percentage of parts correctly sorted into bins; speed; zero damaged parts.	Conveyor belt, parts bins, assembly line.	Jointed robotic arm, pneumatic gripper, conveyor belt speed control.	RGB-D camera, infrared proximity sensors, tactile grip sensors.

Agent Type	Performance Measure (P)	Environment (E)	Actuators (A)	Sensors (S)
Interactive AI Language Tutor	Student mastery test scores, student engagement time, minimized frustration.	Set of students, curriculum database, learning management system.	Display screen (exercises, feedback, hints, motivational voice synthesis).	Keyboard and mouse input, speech recognition microphone, webcam (facial attentiveness).

QUICK RECALL

- PEAS: Performance measure, Environment, Actuators, Sensors.
- Always state Performance measure objectively from the perspective of an external observer.
- Actuators are the mechanisms that affect the world; Sensors are the perception devices.

Environment Classifications & Dimensions

Source: Russell & Norvig 4e, Ch. 1.4, pp. 28–34; slides 46–52

§ 5

The design of an agent program is determined almost entirely by the nature of its task environment. Environments are classified across seven fundamental conceptual dimensions.

Dimension	Dichotomy Definition	Example: Easy vs Hard
Observability	**Fully Observable:** Agent's sensors give complete access to the entire state of the environment at all times. **Partially Observable:** Noise, inaccurate sensors, or hidden state prevents full view.	Chess is fully observable. Poker / Driving is partially observable.
Number of Agents	**Single-Agent:** Agent operates alone. **Multi-Agent:** Environment contains multiple agents. Can be **Competitive** (adversarial) or **Cooperative**.	Crossword puzzle is single-agent. Chess is competitive multi-agent.

Dimension	Dichotomy Definition	Example: Easy vs Hard
Determinism	**Deterministic:** Next state is completely determined by current state and the agent's action. **Stochastic:** Uncertainty or randomness (probabilities) affect outcomes.	Vacuum world without slippage is deterministic. Autonomous driving is stochastic.
Episodic vs Sequential	**Episodic:** Experience is divided into atomic episodes; current decision does not affect future episodes. **Sequential:** Current action affects all future states and rewards (requires planning).	Defective part detection is episodic. Chess and Maze navigation are sequential.
Static vs Dynamic	**Static:** Environment does not change while the agent is deliberating. **Dynamic:** Environment continuously changes while the agent thinks (semidynamic = score drops).	Crossword is static. Taxi driving is dynamic. Timed chess is semidynamic.
Discrete vs Continuous	**Discrete:** Finite, distinct number of states, actions, and percepts. **Continuous:** Smoothly varying values over time (speed, coordinates).	Chess is discrete. Taxi driving is continuous.
Known vs Unknown	**Known:** The laws / physics of the environment are known to the agent. **Unknown:** The agent must explore and learn how the environment works.	Video game with known rules is known. Real world exploration is unknown.

Dimension	Dichotomy Definition	Example: Easy vs Hard
Accessibility (Accessible vs Inaccessible)	**Accessible:** Agent's sensory apparatus gives it full, direct access to the complete environment state (synonymous with fully observable). **Inaccessible:** Sensors are obscured, incomplete, or noisy, withholding full state information.	Grid world with complete sensors is accessible. Underwater autonomous submarine or driving with blinded sensors is inaccessible.

EXAM SUMMARY: THE HARDEST ENVIRONMENT TYPE:

The most difficult task environment for an AI agent is: ***Partially observable, Multi-agent, Stochastic, Sequential, Dynamic, Continuous, and Unknown*** (e.g. Automated Taxi Driving in a foreign city!).

QUICK RECALL

- 7 Dimensions: Observability, Agents, Determinism, Episodic/Sequential, Static/Dynamic, Discrete/Continuous, Known/Unknown.
- Chess: Fully observable, Deterministic, Sequential, Static, Discrete, Known.
- Taxi Driving: Partially observable, Stochastic, Sequential, Dynamic, Continuous, Unknown.

Internal Agent Architectures

Source: Russell & Norvig 4e, Ch. 2.1–2.3, pp. 36–46; AI-Intelligent_Agents-PS.pdf slides 1–20

§ 6

Russell and Norvig define five basic agent architectures of increasing sophistication: ***Simple Reflex***, ***Model-Based Reflex***, ***Goal-Based***, ***Utility-Based***, and ***Learning Agents***.

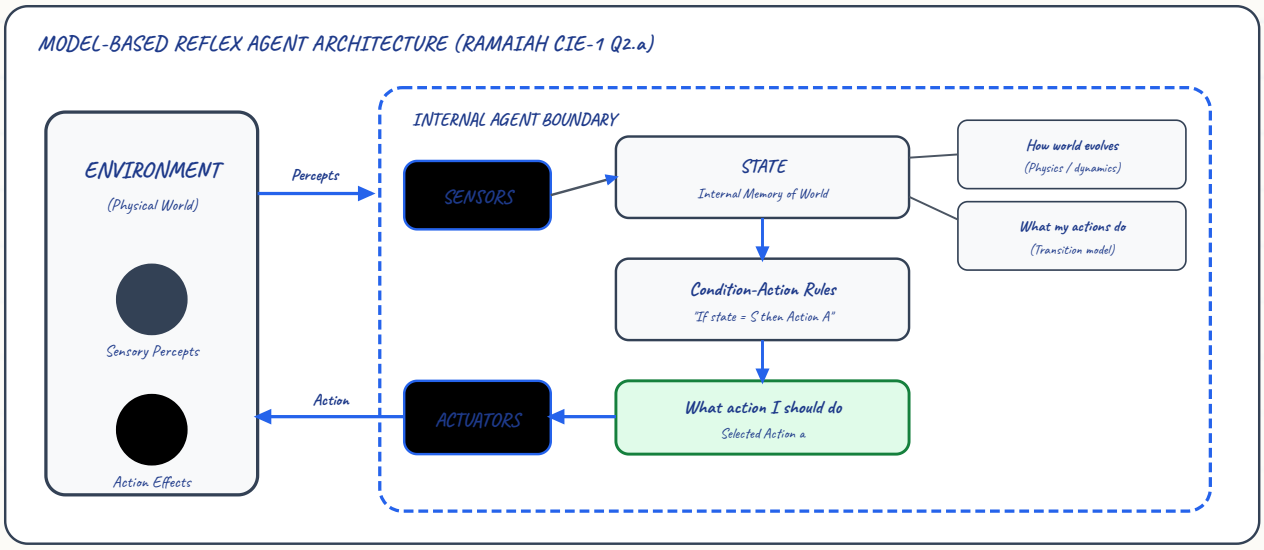


Figure 1.2: Model-Based Reflex Agent Internal Architecture (Ramaiah CIE-1 Q2.a)

The agent maintains an internal state updated via transition model physics ("how world evolves") and sensor model ("what my actions do"), overcoming partial observability.

Agent Architecture	Internal State & Mechanism	Action Selection Rule	Limitation / Failure Mode
1. Simple Reflex Agent	No memory of past percepts. Direct link from current percept to action.	Condition-Action Rules: `if condition then action`	Infinite loops in partially observable environments (e.g. can't escape if state looks identical).
2. Model-Based Reflex Agent	Maintains an ***internal state*** tracking aspects of the unobserved world.	Updates state using transition model ("How world evolves") and sensor model ("How actions affect world").	Lacks explicit future goals; only acts on current beliefs.
3. Goal-Based Agent	Combines internal state with explicit ***Goal information*** describing desirable situations.	Uses ***Search and Planning*** to find sequences of actions that achieve the goal.	Binary satisfaction (goal achieved vs not); cannot trade off between conflicting goals.
4. Utility-Based Agent	Uses an explicit ***Utility function*** mapping states to real	Selects the action that maximizes Expected	Computationally expensive; designing complex multi-attribute

Agent Architecture	Internal State & Mechanism	Action Selection Rule	Limitation / Failure Mode
5. Learning Agent	Divided into 4 functional modules: Critic, Learning Element, Performance Element, Problem Generator.	Learns from feedback to improve performance beyond initial programming.	Requires training samples and exploration-exploitation balance.

The Four Components of a Learning Agent

1

Critic: Evaluates how well the agent is doing against an external, fixed performance standard. Provides feedback to the learning element.

2

Learning Element: Responsible for making improvements. Takes critic feedback and experiments with new behaviors.

3

Performance Element: The traditional agent that selects external actions based on percepts (what we previously called the agent).

4

Problem Generator: Suggests actions that lead to new and informative experiences (exploration), even if they may yield suboptimal short-term rewards.

QUICK RECALL

- Simple reflex: condition-action rules; fails if partially observable.
- Model-based: internal state tracks the world.
- Goal-based: uses search/planning to achieve goal.
- Utility-based: maximizes expected utility (handles trade-offs).
- Learning agent: Critic, Learning Element, Performance Element, Problem Generator.

> **[Interactive Visualizer Online]:** *Interactive Agent Architecture & PEAS Task Environment Simulator. Explore the step-by-step visualizer on the live course notes page.*

Problem Solving by Search & The 5 Components

Source: Russell & Norvig 4e, Ch. 2.3, pp. 44–48; slides 21–35

§ 7

A **problem-solving agent** is a goal-based agent that decides what to do by finding sequences of actions that lead to desirable states. When the correct action cannot be determined immediately, the agent formulates a search problem.

5 FIVE COMPONENTS OF A WELL-DEFINED SEARCH PROBLEM:

1. **Initial State (s_0):** The state that the agent starts in (e.g. $In(Arad)$).
2. **Actions ($Actions(s)$):** A description of the possible actions available to the agent in state s. Given state s, $Actions(s)$ returns the set of legal actions.
3. **Transition Model ($Result(s, a)$):** A description of what each action does. $Result(s, a)$ returns the state that results from doing action a in state s.
4. **Goal Test ($IsGoal(s)$):** Determines whether a given state s is a goal state. Can be explicit (e.g. $s = In(Bucharest)$) or implicit (e.g. checkmate in chess).
5. **Path Cost ($c(s, a, s')$):** A function that assigns a numeric cost to each step. The step cost of taking action a in state s to reach s' is denoted $c(s, a, s')$. The total path cost is the sum of step costs along the path.

A **solution** is an action sequence that leads from the initial state to a goal state. An **optimal solution** is a solution that has the lowest path cost among all solutions.

QUICK RECALL

- 5 Components: Initial State, Actions, Transition Model, Goal Test, Path Cost.
- State Space: set of all reachable states from the initial state.
- Optimal solution = lowest path cost among all solutions.

State Space Formulations for Toy & Real-World Problems

Source: Russell & Norvig 4e, Ch. 2.4, pp. 49–58; slides 36–55

§ 8

A **state space** is the complete set of all states reachable from the initial state by any sequence of actions. State space formulation abstracts away irrelevant physical details to focus on formal problem solving.

CORE PROBLEM FORMULATION · VERIFIED SCRIPT PASS [10 MARKS · EXAM CORE]

Formal State-Space Formulation of the 8-Puzzle

The 8-puzzle consists of a 3×3 board with 8 numbered sliding tiles and one blank space.

1. Formal 5-Component Specification:

- **States:** A specification of the position of each of the 8 numbered tiles and the blank square in the 3×3 grid.
- **Initial State:** Any valid configuration of the board.
- **Actions:** Movements of the blank space: $\{\text{Left}, \text{Right}, \text{Up}, \text{Down}\}$. (Moving the blank is cleaner than moving individual tiles!).
- **Transition Model:** Given state s and blank move a, returns the resulting board with the blank and adjacent tile swapped.
- **Goal Test:** Matches the designated goal configuration (e.g. tiles ordered 1 to 8 with blank at bottom-right or center).
- **Path Cost:** Each step costs 1 (path cost = number of moves).

2. State Space Size & Inversion Parity Analysis:

- Total distinct arrangements of 9 squares $= 9! = \mathbf{362,880}$ states.
- **Inversion Parity Theorem:** The state space is partitioned into **two disjoint graph components of size $9!/2 = 181,440$ states each**. Any legal slide move preserves the even/odd parity of the number of tile inversions. Exactly half the states can reach the goal; the other half are provably impossible!

3. Branching Factor:

- Blank in corner (4 corners): 2 legal moves.
- Blank on edge (4 edges): 3 legal moves.
- Blank in center (1 center): 4 legal moves.
- Average branching factor: $b = \frac{4(2)+4(3)+1(4)}{9} = \frac{8+12+4}{9} = \frac{24}{9} \approx \mathbf{2.67}$.

Formulation Style	State Representation	State Space Size	Search Methodology
Naive Incremental	Any arrangement of 0 to 8 queens on the 64 squares.	$\binom{64}{8} \approx 4.4 \times 10^9$ states	Place queen on any empty square. Massive redundant search.
Clever Incremental	Place one queen per column in columns 1 to k such that no queen is attacked.	$8^8 \approx 1.68 \times 10^7$ states	Place next queen in leftmost empty column without conflict.
Complete-State Formulation	All 8 queens are on the board, exactly one queen per column.	$8! = \mathbf{40,320}$ states	Used by local search (Hill Climbing / Min-Conflicts) to slide queens vertically within their column.

QUICK RECALL

- 8-puzzle: $9! = 362,880$ permutations; $181,440$ reachable states due to inversion parity.
- 8-puzzle average branching factor $b = 24/9 \approx 2.67$.
- 8-queens complete-state formulation reduces search space from 4.4×10^9 to $8! = 40,320$.

One-Page Rapid Revision Sheet

REVISION

Definitions of AI

- Thinking Humanly: cognitive modeling (mind).
- Acting Humanly: Turing Test (behavior).
- Thinking Rationally: formal logic (syllogisms).
- Acting Rationally: ***rational agent*** (modern paradigm).

PEAS & Environments

- PEAS: Performance measure, Environment, Actuators, Sensors.
- 7 Dimensions: Observability, Agents, Determinism, Episodic, Static, Discrete, Known.
- Hardest: Partially observable, Multi-agent, Stochastic, Sequential, Dynamic, Continuous.

Agent Architectures

- Simple Reflex: condition-action rules.
- Model-Based: internal state tracks unseen world.
- Goal-Based: planning and search for goals.
- Utility-Based: maximizes expected utility.
- Learning: Critic, Learning, Performance, Problem Gen.

Problem Formulation

- 5 Components: Initial state, Actions, Transition model, Goal test, Path cost.
- 8-Puzzle: $9!/2 = 181,440$ reachable states; $b \approx 2.67$.
- 8-Queens: $8! = 40,320$ permutations in complete-state.

Important Examination Questions

? QUESTIONS

2-Mark Questions (Short Answer)

Practice Problem:

2M Define an agent and write the agent function mathematically.

Solution:

Answer: An agent is anything that perceives its environment through sensors and acts upon it through actuators. The agent function maps any percept sequence to an action: $f : P^* \to A$.

Practice Problem:

2M Distinguish between rationality and omniscience.

Solution:

Answer: An omniscient agent knows the actual outcome of its actions, which is impossible in real-world environments. Rationality maximizes expected performance based on the agent's percept sequence and prior knowledge to date.

Practice Problem:

2M What are condition-action rules in simple reflex agents?

Solution:

Answer: Condition-action rules are production rules of the form `if condition then action` that map current percepts directly to actions without retaining memory of past percepts.

5-Mark Questions (Medium Length)

Practice Problem:

5M Explain the Turing Test approach to AI and list the capabilities required to pass it.

Solution:

Answer: Proposed by Alan Turing (1950), the Turing Test places a human interrogator in teletype conversation with a human and a computer. If the interrogator cannot tell which is which, the machine passes. Required capabilities:

- Natural Language Processing (NLP) to communicate fluently.
- Knowledge Representation to store what it knows or hears.
- Automated Reasoning to use stored information to answer questions.
- Machine Learning to adapt and detect new patterns.

Practice Problem:

5M Detail the PEAS specification for an automated taxi driver.

Solution:

Answer:

- **P (Performance):** Safety, speed, adherence to traffic laws, passenger comfort, fuel efficiency, profitability.
- **E (Environment):** Urban streets, highway traffic, pedestrians, weather, road construction, passengers.
- **A (Actuators):** Steering wheel, accelerator, brakes, horn, turn signals, display screen.
- **S (Sensors):** Cameras, sonar, LIDAR, GPS, speedometer, engine sensors, accelerometer.

Feature	Artificial Intelligence (AI)	Machine Learning (ML)	Deep Learning (DL)
Scope	Broadest umbrella field: creating intelligent computational behavior.	Subset of AI focused on learning patterns from data without explicit rules.	Specialized subset of ML using deep multi-layered artificial neural networks.
Core Paradigm	Rule-based systems, expert systems, search, logic, and learning.	Statistical algorithms (Linear Regression, SVM, Decision Trees, Naive Bayes).	Representation learning (CNN, RNN, Transformers, ResNets) with automatic feature extraction.
Feature Engineering	Manually hardcoded by knowledge engineers.	Requires human domain experts to craft input feature vectors.	End-to-end automatic hierarchy of feature representations learned from raw inputs.
Data Scale	Can function on zero training data using search trees or logical inference.	Performs well on tabular and medium-sized structured datasets (thousands of rows).	Requires massive scale datasets (millions of parameters) and GPU acceleration.

Authentic Solved Questions from Past Ramaiah CIE & SEE Papers

EXAM ARCHIVE

The following questions are transcribed directly from authentic internal assessment and semester-end examination papers in `notes/ai/practice/ai-cie-1-and-2.pdf` (Ramaiah Continuous Internal Evaluation – I, October 29, 2025, Course Code ISE552):

Ramaiah CIE-1 · October 29, 2025 · Question 1.a [4 Marks]

"Describe the four approaches to defining Artificial Intelligence."

Practice Problem:

Show Complete Step-by-Step Model Answer

Solution:

Historically, researchers have organized definitions of Artificial Intelligence along two fundamental dimensions: **Thought Processes vs. External Behavior**, and **Human Performance vs. Rational (Ideal) Performance** (Russell & Norvig):

Dimension	Human-Centric Approach	Rationality-Centric (Ideal Standard)
Thinking	**1. Thinking Humanly (Cognitive Modeling):** Requires determining how human minds work through psychological experimentation, brain imaging (fMRI), or introspection. Validated when computational execution traces match human cognitive step traces.	**2. Thinking Rationally (Laws of Thought):** Rooted in Aristotelian syllogisms and formal deductive logic. Emphasizes mathematically irrefutable inference ($P \implies Q, P \vdash Q$). Limited by exponential complexity and difficulty encoding informal, uncertain real-world facts.
Acting	**3. Acting Humanly (The Turing Test Approach):** Formulated by Alan Turing (1950). A computer passes if an interrogator cannot distinguish its written answers from a human. Requires NLP, knowledge representation, automated reasoning, and machine learning.	**4. Acting Rationally (The Rational Agent Approach) :** Modern gold standard. A **rational agent** operates to maximize its expected performance measure given available percepts and built-in knowledge. Correct inference is one mechanism, but reflex actions are also rational.

Why Modern AI Focuses on the Rational Agent Approach:

1. It is more general than the "laws of thought" logicist approach, because correct inference is only one way to achieve rationality (e.g., reflexively pulling back from a hot surface achieves rationality without deduction).
2. It is mathematically well-defined and amenable to scientific engineering, whereas human

Ramaiah CIE-1 · October 29, 2025 · Question 2.a [4 Marks]

"Explain Model-Based Reflex Agent with a neat schematic diagram and algorithmic flow."

Practice Problem:

Show Complete Step-by-Step Model Answer

Solution:

1. Architectural Purpose & Motivation:

A **Simple Reflex Agent** fails whenever the task environment is partially observable, because multiple distinct world states can generate identical sensory percepts, causing the agent to fall into infinite loops. A **Model-Based Reflex Agent** overcomes this limitation by maintaining an **internal state** that tracks aspects of the world currently unobservable by sensors.

2. Two Core Internal Knowledge Models:

- **Transition Model ("How the world evolves"):** Knowledge of the physical dynamics of the environment independent of the agent (e.g., if a car is moving ahead, it will continue moving forward in the next time step).
- **Sensor / Effect Model ("What my actions do"):** Knowledge of how the agent's own actions affect the state of the world (e.g., turning the steering wheel left causes the vehicle to rotate counter-clockwise).

3. Algorithmic Flow & Pseudocode:

```
function MODEL-BASED-REFLEX-AGENT(percept) returns an action
  persistent: state, the agent's current conception of the world
state
              model, a description of how the next state depends on
current state and action
              rules, a set of condition-action rules
              action, the most recent action, initially none

  state ← UPDATE-STATE(state, action, percept, model)
  rule ← RULE-MATCH(state, rules)
  action ← rule.ACTION
  return action
```

4. Operational Trace:

1. Sensors receive current percept p_t.
2. UPDATE-STATE combines previous state s_{t-1}, previous action a_{t-1}, current

PEAS Component	Engineering Specification
Performance Measure (P)	Minimize average vehicle wait time and queue lengths; eliminate intersection gridlock; prioritize emergency vehicles (ambulances/fire engines); minimize fuel consumption/emissions; maximize vehicle throughput.
Environment (E)	Multi-lane urban intersection roads, motorized vehicular traffic (cars, buses, two-wheelers), pedestrians at crosswalks, varying weather conditions (rain, fog), time-of-day traffic surges.
Actuators (A)	Traffic signal heads (Red, Yellow, Green cycle timing controllers), variable message electronic sign boards, audible pedestrian crossing tones, motorized emergency barrier gates.
Sensors (S)	High-definition CCTV overhead intersection cameras, in-road inductive loop electromagnetic detectors, infrared/radar vehicle speed sensors, acoustic siren detectors, pedestrian crosswalk pushbuttons.

Ramaiah SEE Archive • Model Question Paper [8 Marks]

"Formulate the 8-Puzzle Problem as a State Space Search Problem. Determine the total permutations, reachable states using the Inversion Parity Theorem, and average branching factor."

Practice Problem:

Show Complete Step-by-Step Model Answer

Solution:

1. Formal 5-Component Problem Formulation:

- **States:** A configuration of the 3×3 grid specifying the exact coordinates of the 8 numbered tiles $\{1, 2, \ldots, 8\}$ and the blank space (denoted 0).
- **Initial State:** Any designated starting arrangement of tiles on the board.
- **Actions:** Operations defined by moving the blank space: $\{\text{Left}, \text{Right}, \text{Up}, \text{Down}\}$. (Moving the blank space simplifies operator design over moving 8 distinct tiles!).
- **Transition Model** $Result(s, a)$**:** Given board state s and blank action a, returns the new board state resulting from swapping the blank with the tile in the target direction.
- **Goal Test:** Checks if the current board matches the target configuration:

$$\text{Goal} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & \text{blank} \end{bmatrix}$$

- **Path Cost:** Each step costs 1. Total path cost equals the total number of moves in the solution path.

2. State Space Size & Inversion Parity Theorem:

- Total unconstrained permutations of 9 tiles:

$$\text{Total Permutations} = 9! = 362,880$$

- An **inversion** occurs when any tile with higher value precedes a tile with lower value in row-major order (ignoring the blank).
- **Theorem:** In an odd-width grid (3×3), horizontal slides do not alter tile ordering; vertical slides displace a tile by exactly 2 positions, changing the inversion count by 0 or ± 2. Thus, **every legal move strictly preserves the even/odd parity of the number of inversions.**

Dimension	Chess with Clock	Autonomous Taxi Driving	Medical Diagnosis System
Observability	**Fully Observable:** Entire board state visible.	**Partially Observable:** Obstructed streets, blind spots.	**Partially Observable:** Unseen internal pathology.
Agents	**Multi-Agent Competitive:** Opponent is an adversary.	**Multi-Agent:** Competitive & cooperative drivers.	**Single-Agent:** Diseases are biological, not adversaries.
Determinism	**Deterministic:** Moves have exact outcomes.	**Stochastic:** Slippery roads, sudden obstacles.	**Stochastic:** Variable biological patient responses.
Episodic / Sequential	**Sequential:** Early moves dictate endgame.	**Sequential:** Steering affects future positions.	**Sequential:** Tests affect subsequent treatments.
Static / Dynamic	**Semidynamic:** Board is static, but clock ticks.	**Dynamic:** Traffic moves while agent computes.	**Static:** Patient status stable during typing.
Discrete / Continuous	**Discrete:** Finite board positions and turns.	**Continuous:** Velocity, steering angle, time.	**Continuous/Discrete:** Vitals vs symptom flags.
Known / Unknown	**Known:** Formal chess rules fixed.	**Unknown:** New roads, changing traffic rules.	**Partially Known:** Medical science vs individual genetics.

Verified High-Yield Practice Problems

PRACTICE

Test your understanding with these exam-caliber problems. All numerical counts and logical derivations are verified against `audit/verify/ai/verify_ai_u1.py`:

PRACTICE 1 · 8-QUEENS STATE SPACE REDUCTION

Compare the state-space sizes of the naive formulation, column-constrained formulation, and complete-state permutation formulation of the 8-Queens problem.

Practice Problem:

Show Step-by-Step Solution

Solution:

1. Naive Unconstrained Formulation: Place 8 queens on any of the 64 squares:

$$\binom{64}{8} = \frac{64!}{8! \times 56!} = \mathbf{4,426,165,368} \text{ states} \approx 4.43 \times 10^9$$

2. Column-Constrained Formulation: Place exactly one queen in each column ($k = 1$ to 8):

$$8^8 = \mathbf{16,777,216} \text{ states} \approx 1.68 \times 10^7$$

3. Complete-State Permutation Formulation: Exactly one queen per column AND per row:

$$8! = \mathbf{40,320} \text{ states}$$

Conclusion: By enforcing row and column constraints directly into the state representation, the search space collapses by a factor of over $\mathbf{100,000}\times$!

PRACTICE 2 · 8-PUZZLE REACHABILITY THEOREM

Can the start state $S = [1, 2, 3, 4, 5, 6, 8, 7, 0]$ reach the standard goal state $G = [1, 2, 3, 4, 5, 6, 7, 8, 0]$? Justify mathematically.

Practice Problem:

Show Step-by-Step Solution

Solution:

Step 1: Compute inversions of Goal state G:

In $G = [1, 2, 3, 4, 5, 6, 7, 8]$, all numbers are in increasing order. Number of inversions $= 0$ (Even parity).

Step 2: Compute inversions of Start state S:

Writing S in row-major order (excluding blank 0): $1, 2, 3, 4, 5, 6, 8, 7$.

- For tile 8: followed by tile 7 $(8 > 7) \rightarrow 1$ inversion.
- All other tile pairs are in natural order.
- Total inversions in $S = \mathbf{1}$ (Odd parity).

Step 3: Parity Invariance:

Legal tile slides can only alter the inversion count by an even number (0 or ± 2).

Therefore, an odd-parity state can **never** reach an even-parity goal.

Final Answer: Mathematically unreachable (impossible!).

PRACTICE 3 · BRANCHING FACTOR DERIVATION

Derive the average branching factor of the 8-puzzle and explain why blank moves simplify problem formulation.

Practice Problem:

Show Step-by-Step Solution

Solution:

A 3×3 board contains 9 possible positions for the blank space:

- **4 Corners:** Blank can move in 2 directions (along adjacent edges) $\rightarrow 4 \times 2 = 8$ legal moves.
- **4 Edges:** Blank can move in 3 directions $\rightarrow 4 \times 3 = 12$ legal moves.
- **1 Center:** Blank can move in 4 directions $\rightarrow 1 \times 4 = 4$ legal moves.

Assuming the blank is equally likely to occupy any square across a random search trajectory:

$$b_{\text{avg}} = \frac{4(2) + 4(3) + 1(4)}{9} = \frac{8 + 12 + 4}{9} = \frac{24}{9} = \mathbf{2.67}$$

Formulation Advantage: Formulating actions as "move blank" $\{\text{Left}, \text{Right}, \text{Up}, \text{Down}\}$ creates an action set of size 4, whereas formulating actions as "move tile k" requires tracking which of the 8 tiles is adjacent to the blank, creating up to 32 cumbersome rules!

PRACTICE 4 · ENVIRONMENT CLASSIFICATION FOR POKER

Classify the task environment of playing Texas Hold'em Poker across the 7 environmental dimensions.

Practice Problem:

Show Step-by-Step Solution

Solution:

1. **Observability:** Partially Observable — opponents' hole cards are concealed.
2. **Agents:** Multi-Agent Competitive — opponents act to maximize their chip count at other players' expense.
3. **Determinism:** Stochastic — card dealing introduces irreducible probability distributions.
4. **Episodic vs. Sequential:** Sequential — chip stacks, bluff reputations, and betting actions carry over across rounds.
5. **Static vs. Dynamic:** Static (or Semidynamic in online timed poker) — hole cards and table pot do not mutate while the player is calculating pot odds.
6. **Discrete vs. Continuous:** Discrete — finite deck of 52 cards, discrete betting limits, discrete betting actions (Fold, Call, Raise).
7. **Known vs. Unknown:** Known — formal rules of poker and hand rankings are fully known to all players.

Current Percept [Location, Status]	Rational Action	Justification
$[A, \text{Dirty}]$	**Suck**	Cleans dirty location A.
$[A, \text{Clean}]$	**Right**	Moves to explore location B.
$[B, \text{Dirty}]$	**Suck**	Cleans dirty location B.
$[B, \text{Clean}]$	**Left**	Moves to explore location A.

PRACTICE 6 · RATIONALITY VS OMNISCIENCE SCENARIO

An autonomous vehicle travels safely at 40 km/h on an empty road. A boulder suddenly dislodges from an unmonitored cliff face and crushes the car. Was the vehicle agent irrational? Explain.

Practice Problem:

Show Step-by-Step Solution

Solution:

Answer: No, the agent was completely rational!

Engineering Justification:

- An **omniscient** agent knows the actual future outcome of its actions with supernatural precision. An omniscient car would have stopped before the cliff because it "knew" the boulder would fall. However, omniscience is physically impossible in real-world stochastic environments.
- A **rational** agent maximizes expected performance based on its built-in knowledge, sensory percepts, and past experience. Traveling at the legal speed limit of 40 km/h with no prior percept of overhead cliff instability is the action that maximizes expected utility.
- Rationality is judged on the quality of decision-making given available information ex-ante, whereas perfection in hindsight is the criterion of omniscience. Therefore, an unexpected catastrophic event does not imply irrationality.